

SQL Skills

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1  -- Write a SQL query to create WORKER Table
2  CREATE TABLE worker(
3  worker_id INT NOT NULL PRIMARY KEY,
4  first_name VARCHAR(25),
5  last_name VARCHAR(25),
6  salary INT,
7  joining_date DATETIME,
8  department VARCHAR(25));
9
10 -- Write a SQL Query to insert values in WORKER Table
11 INSERT INTO worker (
12 worker_id, first_name, last_name, salary, joining_date, department) VALUES
13 (1, 'Siddharth', 'Singh', 80000, '2019-03-20 09:00:00', 'HR'),
14 (2, 'Lavesh', 'Ahir', 300000, '2019-07-11 09:00:00', 'Admin'),
15 (3, 'Abhishek', 'Midha', 500000, '2019-03-20 09:00:00', 'HR'),
16 (4, 'Rahul', 'Mahar', 200000, '2019-03-20 09:00:00', 'Admin'),
17 (5, 'Saurabh', 'Madavi', 90000, '2019-07-11 09:00:00', 'Admin'),
18 (6, 'Aman', 'Nain', 75000, '2019-07-11 09:00:00', 'Account'),
19 (7, 'Vaibhav', 'Varshney', 100000, '2019-02-20 09:00:00', 'Account'),
20 (8, 'Farhaan', 'Majied', 500000, '2019-05-11 09:00:00', 'Admin');
21
22 -- Write a SQL Query to create table Title which has WORKER_REF_ID
23 -- as foreign key
24 CREATE TABLE title(
25 worker_ref_id INT NOT NULL PRIMARY KEY,
26 worker_title VARCHAR(25),
27 affected_from DATETIME,
28 FOREIGN KEY (worker_ref_id)
29 REFERENCES worker(worker_id)
30 ON DELETE CASCADE);
31
32 -- Write a SQL Query to insert values in TITLE Table
33 INSERT INTO title(
34 worker_ref_id, worker_title, affected_from) VALUES
35 (1, 'Manager', '2021-02-20 00:00:00'),
36 (2, 'Executive', '2021-06-11 00:00:00'),
37 (8, 'Executive', '2021-06-11 00:00:00'),
38 (5, 'Manager', '2021-06-11 00:00:00'),
39 (4, 'Asst. Manager', '2021-06-11 00:00:00'),
40 (7, 'Executive', '2021-06-11 00:00:00'),
41 (6, 'Lead', '2021-06-11 00:00:00'),
42 (3, 'Lead', '2021-06-11 00:00:00');
43
44 -- Write a SQL Query to insert values in JOB GRADE Table
45 CREATE TABLE job_grade(
46 grade_level VARCHAR(25),
47 lowest_sal INT,
48 highest_sal INT);
49
50 -- Write a SQL Query to insert values in JOB GRADE Table
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51 INSERT INTO job_grade (  
52 grade_level, lowest_sal, highest_sal) VALUES  
53 ('A', 10000, 75999),  
54 ('B', 76000, 80999),  
55 ('C', 81000, 99999),  
56 ('D', 100000, 199999),  
57 ('E', 200000, 299999),  
58 ('F', 300000, 600000);  
59  
60 -- Write a SQL query to fetch "FIRST_NAME" from Worker table in upper case.  
61 SELECT upper(first_name)  
62 FROM worker;  
63  
64 -- Write an SQL query to fetch unique values of DEPARTMENT from Worker table.  
65 SELECT DISTINCT department  
66 FROM worker;  
67  
68 -- Write a SQL query to print the FIRST_NAME from Worker table after replacing 'a'  
69 with 'A'  
70 SELECT REPLACE(first_name, 'a', 'A')  
71 FROM worker;  
72  
73 -- Write a SQL query to print the FIRST_NAME and LAST_NAME  
74 -- from Worker table into a single column COMPLETE_NAME. A  
75 -- space char should separate them.  
76 SELECT CONCAT(first_name, ' ', last_name)  
77 FROM worker;  
78  
79 -- Write a SQL query to print all Worker details from the Worker table  
80 -- order by FIRST_NAME Ascending.  
81 SELECT *  
82 FROM worker  
83 ORDER BY first_name ASC;  
84  
85 -- Write a SQL query to print all Worker details from the  
86 -- Worker table order by FIRST_NAME Ascending and  
87 -- DEPARTMENT Descending.  
88 SELECT *  
89 FROM worker  
90 ORDER BY first_name ASC, department DESC;  
91  
92 -- Write a SQL query to print details for Workers with the first  
93 -- name as "Rahul" and "Lavesh" from Worker table.  
94 SELECT *  
95 FROM worker  
96 WHERE first_name = 'Rahul' OR first_name = 'Lavesh';  
97  
98 -- Write a SQL query to print details of workers excluding first  
99 -- names, "Rahul" and "Lavesh" from Worker table.  
100 SELECT *  
101 FROM worker  
102 WHERE first_name NOT IN ('Rahul', 'Lavesh');  
103  
104 -- Write a SQL query to print details of the Workers whose
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104 -- FIRST_NAME starts with 'S'
105 SELECT *
106 FROM worker
107 WHERE first_name LIKE 'S%';
108
109 -- Write a SQL query to print details of the Workers whose
110 -- FIRST_NAME contains 'a'
111 SELECT *
112 FROM worker
113 WHERE first_name LIKE '%a%';
114
115 -- Write a SQL query to print details of the Workers whose
116 -- FIRST_NAME ends with 'n'.
117 SELECT *
118 FROM worker
119 WHERE first_name LIKE '%n';
120
121 -- Write a SQL query to print details of the Workers whose
122 -- FIRST_NAME ends with 'l' and contains five alphabets.
123 SELECT *
124 FROM worker
125 WHERE first_name LIKE '____l';
126
127 -- Write a SQL query to print details of the Workers whose SALARY lies
128 -- between 100000 and 500000.
129 SELECT *
130 FROM worker
131 WHERE salary BETWEEN 100000 AND 500000;
132
133 -- Write a SQL query to print details of the Workers who have
134 -- joined in Mar'2019.
135 SELECT *
136 FROM worker
137 WHERE year(joining_date) = 2019 and month(joining_date) = 3;
138
139 -- Write a SQL query to show the current date and time.
140 SELECT CURDATE(); -- Returns the current date
141 SELECT now(); -- returns the current date and time
142
143 -- Write a SQL query to fetch the count of employees working in the
144 -- department 'Admin'
145 SELECT COUNT(*)
146 FROM worker
147 WHERE department = 'Admin';
148
149 -- Write a SQL query to fetch the no. of workers for each
150 -- department in the descending order.
151 SELECT department, count(worker_id) no_of_workers
152 FROM worker
153 GROUP BY 1
154 ORDER BY no_of_workers DESC;
155
156 -- Write a SQL query to fetch departments along with the total
157 -- salaries paid for each of them.
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158 SELECT department, SUM(salary) AS total_salary
159 FROM worker
160 GROUP BY 1;
161
162 -- Write a SQL query to show all departments along with the
163 -- number of people in there.
164 SELECT department, count(department) AS 'number of workers'
165 from worker
166 GROUP BY 1;
167
168 -- Write a SQL query to fetch the departments that have less than five
169 -- people in it.
170 SELECT department, count(worker_id) no_of_workers
171 FROM worker
172 GROUP BY 1
173 HAVING count(worker_id) < 5;
174
175 -- Write a SQL query to show one row twice in results from a
176 -- table with department 'HR'.
177 SELECT first_name, department
178 FROM worker AS w
179 WHERE w.department = 'HR'
180 UNION ALL
181 SELECT first_name, department
182 FROM worker AS w1
183 WHERE w1.department = 'HR';
184
185 -- Write a SQL query to fetch the names of workers who earn
186 -- the highest salary.
187 SELECT first_name, salary
188 FROM worker
189 WHERE salary = (
190 SELECT max(salary)
191 FROM worker);
192
193 -- Write a SQL query to show the second highest salary from a
194 -- table.
195 SELECT max(salary)
196 FROM worker
197 WHERE salary NOT IN (
198 SELECT max(salary)
199 FROM worker);
200
201 -- Write a SQL query to show the top n (say 5) records of a table
202 SELECT *
203 FROM worker
204 ORDER BY salary DESC
205 LIMIT 5;
206
207 -- Write a SQL query to determine the nth (say n=3) highest
208 -- salary from a table.
209 SELECT salary
210 FROM worker
211 ORDER BY salary DESC
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212 LIMIT 2,1;
213
214 -- Write a SQL query to print details of the Workers who are also
215 -- Managers.
216 SELECT DISTINCT w.first_name, t.worker_title
217 FROM worker w
218 INNER JOIN title t
219 ON w.worker_id = t.worker_ref_id
220 AND t.worker_title IN ('Manager');
221
222 -- Write a SQL query to find the first name, last name, salary,
223 -- and job grade for all employees.
224 SELECT w.first_name, w.last_name, w.salary, j.grade_level
225 FROM worker w
226 JOIN job_grade j
227 ON w.salary
228 BETWEEN j.lowest_sal AND j.highest_sal;
229
230 -- Write a SQL query to compute the average salary of
231 -- Workers for each job title.
232 SELECT worker_title, avg(salary)
233 FROM worker
234 JOIN title
235 ON worker.worker_id = title.worker_ref_id
236 GROUP BY worker_title;
237
238 -- Write a SQL query to print the name of employees having the
239 -- highest salary in each department
240 SELECT t.department, t.first_name, t.Salary
241 FROM (
242 SELECT max(Salary) as TotalSalary, department
243 FROM Worker
244 group by department) as TempNew
245 INNER JOIN Worker t
246 ON TempNew.department = t.department
247 and TempNew.TotalSalary = t.Salary;
248
249 -- Write a SQL query to create a view for those Worker
250 -- belonging to the Department 'HR'
251 CREATE VIEW hrstaff
252 AS SELECT *
253 FROM Worker
254 WHERE department = 'HR';
255
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